4.2 Subspaces

A subset W of a vector space V is called a subspace of V if W is itself a vector space under the addition and scalar multiplication defined on V.

In general, to show that a nonempty set *W* with two operations is a vector space one must verify the ten vector space axioms. However, if *W* is a subspace of a vector space *V*, then certain axioms need not be verified because they are "inherited" from *V*. For example, it is not necessary to verify that $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ holds in *W* because it holds for all vectors in *V* including those in *W*. On the other hand, it *is* necessary to verify that *W* is closed under addition and scalar multiplication since it is possible that adding two vectors in *W* or multiplying a vector in *W* by a scalar produces a vector in *V* that is outside of *W*.

Those axioms that are *not* inherited by *W* are:
Axiom 1—Closure of *W* under addition
Axiom 4—Existence of a zero vector in *W*
Axiom 5—Existence of a negative in *W* for every vector in *W*
Axiom 6—Closure of *W* under scalar multiplication
so these must be verified to prove that it is a subspace of *V*. However, the following theorem shows that if Axiom 1 and Axiom 6 hold in *W*, then Axioms 4 and 5 hold in *W* as a consequence and hence need not be verified.

THEOREM 4.2.1

If W is the subset of a vector space V, then W is called subspace if and only if the following conditions hold:

a) (Closed under addition) If $\vec{u}$ and $\vec{v}$ are vectors in W, then $\vec{u} + \vec{v}$ is in W.
b) (Closed under scalar multiplication) If k is any scalar and $\vec{u}$ is any vector in W then $k\vec{u}$ is in W.

Or in other words, W is subspace of V if and only if it is closed under addition and scalar multiplication.

Note that every vector space has at least two subspaces, itself and its zero subspace.

Example 1: If V is any vector space and if $W = \{\vec{0}\}$ is the subset of V that consists of the zero vector only, then W is closed under addition and scalar multiplication since

$$\vec{0}+\vec{0}=\vec{0}$$

$$k\vec{0}=\vec{0},\ \text{for any scalar k}$$

We call W the zero subspace of V.

Example 2:

$$W=\{(x,y);x\geq 0,y\geq 0\}$$

Prove and disprove W is subspace of R^2.

Solution:

1. W is closed under addition, because

Let, $\vec{u}=(x,y),\vec{v}=(x',y')\in W$ $\quad (x,y\geq 0\ and\ x',y'\geq 0)$

$\vec{u}+\vec{v}=(x+x',y+y')\in W$ $\quad (x+x'\geq 0\ \ and\ y+y'\geq 0)$

2. W is not closed under scalar multiplication, because if

$$\vec{u}=(1,1)\in W$$

But $-\vec{u}=(-1,-1)\notin W$
So W is not subspace of V.

Example 3:

Which of the following subset of R^2 with usual operations of addition and scalar multiplication are subspaces?

a) $W_1=\{(x,y);x\geq 0\}$
b) $W_2=\{(0,y);y\in R\}$

Solution:

a) W_1 is closed under addition but not closed under scalar multiplication, because

$$\text{Let}\quad \vec{u}=(2,3)\in W$$

$$\text{and}\quad (-2)\vec{u}=-2(2,3)=(-4,-6)\notin W$$

b) $W_3 = \{(0, y); y \in R\}$

As $W_3 = y -$ axis in R^2. To see whether W_3 is subspace , Let

$$\vec{u} = (0, u_1), \vec{v} = (0, v_1) \in W_3$$

$$\vec{u} + \vec{v} = (0, u_1 + v_1) \in W_3,$$

W_3 is closed under addition.

Let $\vec{u} = (0, u_1)$, k is any scalar then

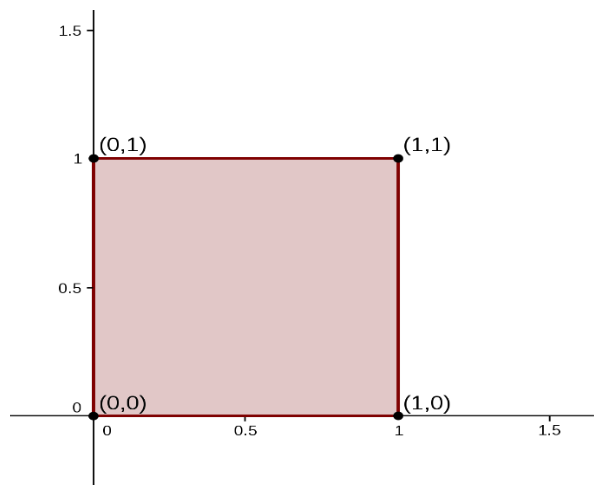
$$k\vec{u} = (0, ku_1) \in W_3$$

W_3 is closed under scalar multiplication.

$\therefore W_3$ is subspace of R^2.

Example 4.

Consider the unit square shown in the accompanying figure. Let W be the set of the form $\begin{bmatrix} x \\ y \end{bmatrix}$, where $0 \le x \le 1, 0 \le y \le 1$. That is W is the set of all vectors whose tail is at origin and whose head is a point inside or on the square. Is W a subspace of R^2? Explain.



Solution:

Let $\vec{u} = (0,1), \vec{v} = (1,1) \in W$.

Then $\vec{u} + \vec{v} = (1,2) \notin W$.

W is not closed under addition.

W is not subspace of R^2.

Example 5.

Which of the given subsets of the vector space M_{23} of all 2×3 matrices are subspace? The set of all matrices of the form?

a) $W_1 = \left\{ \begin{bmatrix} a & b & c \\ d & 0 & 0 \end{bmatrix}; where\ b = a + c \right\}$

b) $W_2 = \left\{ \begin{bmatrix} a & b & c \\ d & 0 & 0 \end{bmatrix}; where\ \ c > 0 \right\}$

c) $W_3 = \left\{ \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}; where\ a = -2c, f = 2c + d \right\}$

Solution:

Do it your self

Exercise 4.2

Q1. Which of the following subset of R^3 with usual operations of addition and scalar multiplication are subspaces?

a) W_1 = All vectors of the form $(a, 0, 0)$
b) W_2 = All vectors of the form $(a, 1, 1)$
c) W_3 = All vectors of the form (a, b, c) where $b = a + c$.
d) W_4 = All vectors of the form (a, b, c) where $b = a + c + 1$.
e) W_5 = All vectors of the form $(a, b, 0)$.

Solution:

a) $W_1 = \{(a, 0, 0); a \in R\}$

1. W_1 is closed under addition and scalar multiplication as:

$$\vec{u} = (a_1, 0, 0) \in W_1$$
$$\vec{v} = (a_2, 0, 0) \in W_1$$
$$\vec{u} + \vec{v} = (a_1 + a_2, 0, 0) \in W_1$$

2. Let $\vec{u} = (a_1, 0, 0) \in W_1$ and k is any scalar, then

$$k\vec{u} = (ka_1, 0, 0) \in W_1$$

W_1 is subspace of R^3.

b) $W_2 = \{(a, 1, 1); a \in R\}$

Let's check the first property, i.e. W_2 is closed under addition. For

$$\vec{u} = (a_1, 1, 1) \in W_2$$
$$\vec{v} = (a_2, 1, 1) \in W_2$$
$$\vec{u} + \vec{v} = (a_1 + a_2, 2,2) \notin W_2$$

W_2 is not subspace of R^3.

c) $W_3 = \{(a, b, c); b = a + c; a, b, c \in R\}$

1. W_1 is closed under addition and scalar multiplication as:

$$\vec{u} = (a_1, b_1, c_1) = (a_1, a_1 + c_1, c_1) \in W_3$$
$$\vec{v} = (a_2, b_2, c_2) = (a_2, a_2 + c_2, c_2) \in W_3$$

Then $\vec{u} + \vec{v} = (a_1 + a_2, (a_1 + c_1) + (a_2 + c_2), c_1 + c_2)$

$$\vec{u} + \vec{v} = (a_1 + a_2, a_1 + a_2 + c_1 + c_2, c_1 + c_2) \in W_3$$

So W_3 is closed under addition.

2. Let $\vec{u} = (a_1, a_1 + c_1, c_1) \in W_3$ and k is any scalar, then

$$k\vec{u} = k(a_1, a_1 + c_1, c_1) = (ka_1, k(a_1 + c_1), kc_1)$$
$$= (ka_1, ka_1 + kc_1, kc_1) \in W_3$$

W_3 is closed under scalar multiplication.

So, W_3 is subspace.

d) $W_4 =$ All vectors of the form (a, b, c) where $b = a + c + 1$.

$$\vec{u} = (a_1, b_1, c_1) = (a_1, a_1 + c_1 + 1, c_1) \in W_4$$
$$\vec{v} = (a_2, b_2, c_2) = (a_2, a_2 + c_2 + 1, c_2) \in W_4$$

Then $\vec{u} + \vec{v} = (a_1 + a_2, (a_1 + c_1) + (a_2 + c_2) + 2, c_1 + c_2)$

$$\vec{u} + \vec{v} = (a_1 + a_2, a_1 + a_2 + c_1 + c_2 + 2, c_1 + c_2) \notin W_4$$

So W_4 is not closed under addition.

So W_4 is not subspace of R^2.

e) $W_5 =$ All vectors of the form $(a, b, 0)$.

Solution:

2. Use Theorem 4.2.1 to determine which of the following are subspaces of M_{nn}.

(a) The set of all diagonal $n \times n$ matrices.

(b) The set of all $n \times n$ matrices A such that $\det(A) = 0$.

(c) The set of all $n \times n$ matrices A such that $\operatorname{tr}(A) = 0$.

(d) The set of all symmetric $n \times n$ matrices.

(e) The set of all $n \times n$ matrices A such that $A^T = -A$.

(f) The set of all $n \times n$ matrices A for which $A\mathbf{x} = 0$ has only the trivial solution.

(g) The set of all $n \times n$ matrices A such that $AB = BA$ for some fixed $n \times n$ matrix B.

Polynomials of degree ≤ n:

$$P(t) = a_n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0; \qquad a_n, a_{n-1}, a_n, \ldots, a_1, a_0 \in R \text{ and } a_n \neq 0$$

Let P_n be the set of all polynomials of degree n, form vector space under addition and scalar multiplication defined by:

$$p(t) = a_n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0$$

$$q(t) = b_n t^n + b_{n-1} t^{n-1} + \cdots + b_1 t + b_0$$

$$p(t) + q(t) = (a_n + b_n) t^n + (a_{n-1} + b_{n-1}) t^{n-1} + \cdots + (a_1 + b_1) t + (a_0 + b_0)$$

If k is any scalar then

$$kp(t) = k(a_n) t^n + k(a_{n-1}) t^{n-1} + \cdots + k(a_1) t + k(a_0)$$

Then P_n is vector space.

3. Use Theorem 4.2.1 to determine which of the following are subspaces of P_3.

(a) All polynomials $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ for which $a_0 = 0$.

(b) All polynomials $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ for which $a_0 + a_1 + a_2 + a_3 = 0$.

(c) All polynomials of the form $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ in which a_0, a_1, a_2, and a_3 are integers.

(d) All polynomials of the form $a_0 + a_1 x$, where a_0 and a_1 are real numbers.